

Roll Number -	
Thapar Institute of Engineering and Technology, Patiala Department of Computer Science and Engineering End Semester Examination (EST-May, 26)	
BE- I st Year: Semester –II	Course Code: UAI201 Course Name: Programming Essentials for AI and ML
23 rd May, 2026	2-5 PM
Time: 3 hours; Marks: 40	Faculty: Dr. Palika Chopra (1X21-1X24), Dr. Ashima Khosla (1X11-1X14)
Important: Solve in order, new problem, new page, if not sure, leave appropriate pages in between. Plan on the last page to avoid cutting. No partial credit or rechecking for the mess. Show all the details and box the answer. One mark to follow the order.	

QNo	Attempt any six questions out of seven. Only the first six attempted questions will be checked and marked.	MM	CO, BL
Q1 (a)	Differentiate between XML and JSON in <u>table format only</u> by writing any three differences between them.	3	CO4, BL2
(b)	A smart city control room wants to store and manage air quality information of a city area using XML and Python. The system should be able to create XML data, read the XML file, extract area details, and modify air quality information. Accordingly, fill in the blanks in the following Python program. <pre> 1. import xml.etree.ElementTree as ET 2. # Create root element 3. city = ET._____("smart_city") 4. # Create area element 5. area = ET._____ (city, "area") 6. ET.SubElement(area, "name").text = "Sector 21" 7. ET.SubElement(area, "air_quality").text = "Good" 8. # Create XML tree and write file 9. tree = ET._____ (city) 10. tree._____ ("smart_city.xml") 11. print("XML file created successfully") 12. # Read XML file 13. tree = ET._____ ("smart_city.xml") 14. root = tree._____ () 15. print("\nArea Details:") 16. for area in root: 17. name = area._____ ("name").text 18. air = area.find("air_quality").text 19. print(name, air) 20. # Modify XML data 21. for area in root: 22. name = area.find("name").text 23. if name == "Sector 21": 24. area.find("air_quality").text = "Poor" 25. tree.write("smart_city.xml") 26. print("\nAir quality updated successfully") </pre> Note: Do not repeat the code for fill-in-the-blank questions. Only provide the answers with their corresponding line numbers	3.5	CO4, BL3
Q2	A forest department wants to monitor forest growth. For each tree, the system should take input of 6 iterations from user in the mentioned variables: Tree Name, Height of Tree, Age of Tree. The system should: ➤ stop data entry if the user enters "stop" as tree name, ➤ skip invalid tree heights (valid height >2), ➤ classify trees based on height, ➤ print & count each category of trees and total valid trees . Tree Height Classification Height < 5 m → Small Tree Height 5–10 m → Medium Tree Height > 10 m → Tall Tree Write a Python program to implement the above scenario. Note: Do not use lists, tuples, sets, or string slicing.	6.5	CO2, BL3
Q3	Write a single-line code for: (Note: Write only the solution by mentioning the question number. Do not repeat the question.) - creating a list, - finding the length of a list, - adding elements to a set, - displaying elements of a tuple, - creating a dictionary, and - finding the maximum element from a list.	6.5	CO3, BL2 P.T.O

Q4(a)	Differentiate between SOAP and REST in table format only by writing any three differences between them.		3.5	CO4, BL2																														
(b)	A college wants to store and manage student information in JSON format using Python. The system should be able to convert Python dictionary data into JSON format, write JSON data into a file, and read the JSON file. Accordingly, fill in the blanks in the following Python program. <pre>1. import json 2. # Create Python dictionary 3. student = { 4. "name": "Aman", 5. "rollno": 21, 6. "cgpa": 8.7, 7. "phonenumber": "9876543210" 8. }</pre>	<pre>9. # Convert Python dictionary to JSON string 10. json_object= json._____ (student, indent=4) 11. # Write JSON string to file 12. with open("student.json", "w") as outfile: 13. outfile._____ (json_object) 14. print("JSON file created successfully") 15. # Read JSON file 16. with open("student.json", "r") as infile: 17. data = json._____ (infile) 18. print("Student Data:") 19. print(data)</pre> Note: Do not repeat the code for fill-in-the-blank questions. Only provide the answers with their corresponding line numbers	3	CO4, BL3																														
Q5	A college wants to analyze student marks out of 100 using Pandas. Some values in the dataset are missing. Perform the following tasks on the given dataset: - Create a DataFrame - Find null values in the dataset - Treat missing values using a suitable method - Calculate the marks percentage for each student and display students scoring above 80% - Sort data according to Percentage - Plot a bar graph for subject-wise marks	<table><tr><th>Roll No</th><th>Name</th><th>Python</th><th>AI</th><th>Data Science</th></tr><tr><td>101</td><td>Aman</td><td>85</td><td>78</td><td>90</td></tr><tr><td>102</td><td>Riya</td><td>92</td><td>NULL</td><td>95</td></tr><tr><td>103</td><td>Karan</td><td>70</td><td>75</td><td>NULL</td></tr><tr><td>104</td><td>Simran</td><td>88</td><td>91</td><td>85</td></tr><tr><td>105</td><td>Arjun</td><td>NULL</td><td>65</td><td>70</td></tr></table> Note: Write only the solution by mentioning the question number. Do not repeat the question.	Roll No	Name	Python	AI	Data Science	101	Aman	85	78	90	102	Riya	92	NULL	95	103	Karan	70	75	NULL	104	Simran	88	91	85	105	Arjun	NULL	65	70	6.5	CO4, BL2
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Q6	A palindrome number is a number that remains the same when its digits are reversed, such as 121 and 1331. Write a Python program to check whether a given number is a palindrome number or not. Input: 121 Output: Palindrome Number Note: Do not use lists, tuples, sets, or string slicing.		6.5	CO1, BL2																														
Q7(a)	A software company is developing a secure banking application in Python where unexpected runtime errors must be handled efficiently to prevent system failure. The application should validate user input, handle division errors, manage multiple exceptions, execute cleanup operations using a finally block, and raise a custom exception for invalid withdrawal amount. Write a Python program to implement the above scenario.		3	CO3, BL3																														
(b)	A retail company wants to store product sales records in a text file. The system should accept product details, skip invalid records, write and read file data, calculate total sales, count valid records, use tell() and seek() functions, and handle file-related exceptions. Write a Python program to implement the above scenario.		3.5																															

*****All the Best*****